

SIMPLIFYING RADICALS NOTES

PARTS OF A RADICAL	The n^{th} root of a number, a, can be written as the radical expression $\sqrt[n]{a}$ $\sqrt[n]{a}$ *If there is no index, it is assumed that _____		
	If a radical has MORE THAN ONE ROOT, the radical sign indicates only the principal, or positive, root.		
PERFECT SQUARE	List the first 12 perfect squares:		
PERFECT SQUARE ROOTS	1. $\sqrt{81}$	2. $\sqrt{484}$	3. $\sqrt{64}$
	4. $18\sqrt{4}$	5. $3\sqrt{256}$	6. $-5\sqrt{121}$
	7. $\sqrt{\frac{1}{16}}$	8. $\sqrt{\frac{9}{100}}$	9. $\sqrt{\frac{25}{49}}$
SIMPLIFYING NON-PERFECT SQUARE ROOTS	1.	Find the largest perfect square that the radical is divisible by. Break down the radical using this number.	
	2.	Take the square root of the perfect square. Take it out of the radical.	
	3.	Leave the "leftover" under the radical symbol.	
	10. $\sqrt{24}$	11. $\sqrt{48}$	12. $\sqrt{72}$
	13. $\sqrt{63}$	14. $\sqrt{90}$	15. $\sqrt{175}$

16. $\sqrt{162}$	17. $\sqrt{245}$	18. $\sqrt{343}$
19. $\sqrt{117}$	20. $\sqrt{28}$	21. $\sqrt{450}$
22. $5\sqrt{40}$	23. $4\sqrt{147}$	